Capstone presentations and your path from here

Can you explain what you built, what it can and cannot do, and what you will learn next?

AI and Data Science Professional | Session 24 of 24 | Module F: Ship it and present

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Today in one view: what you will leave with

By the end of today you can



Tell a result as a story: the answer, the evidence beside a baseline model, and the limits

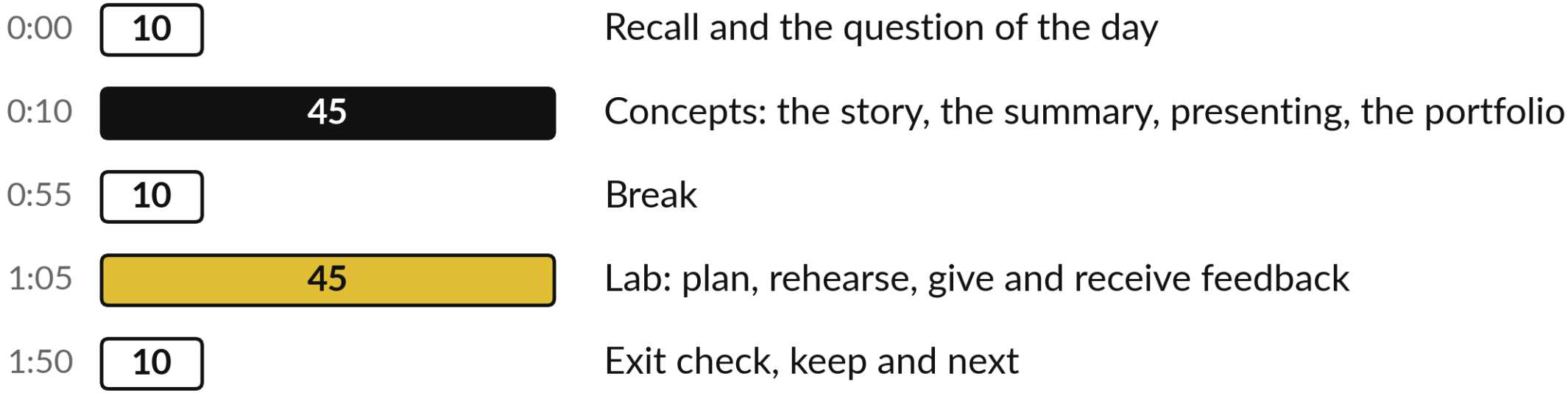


Write an executive summary, present to a manager and answer questions on method and risk



Name what this course did not cover and choose your next free courses and a portfolio plan

Your two hours: concepts first, then a rehearsal lab



Ideas first. Then you plan and rehearse your talk in a sheet. Then the checks and what you keep.

Three questions from last time

1. What is the difference between data drift and concept drift?

2. A rule never uses gender, yet it approves 70.3% of women and 52.4% of men. What is going on, and what do you measure?

3. Name two things that never go into a public GitHub repository.



Answer aloud first. Then we check together.

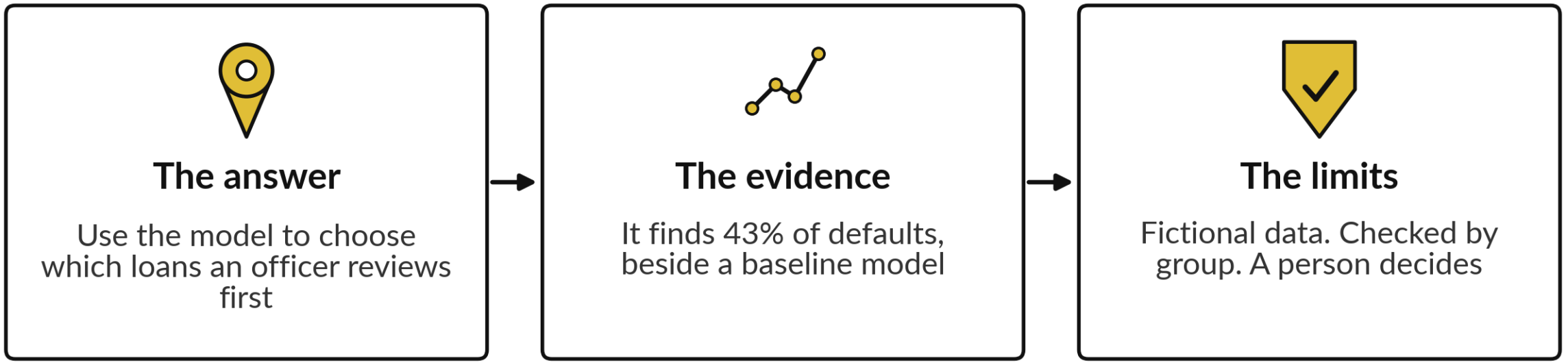
Answers to the three questions from Session 23

Question	Short answer
What is the difference between data drift and concept drift?	In data drift the inputs change after training. In concept drift the link between the inputs and the outcome changes.
A rule never uses gender, yet it approves 70.3% of women and 52.4% of men. What is going on, and what do you measure?	A proxy variable (occupation) carries the information. Measure approval and error rates by group and the disparate impact ratio, here 0.745.
Name two things that never go into a public GitHub repository.	Personal data such as names, phone numbers or loan records of real people, and passwords, keys or tokens.

Words we use today

	Word	What it means
	**Capstone**	The larger project that ends a course. Yours is a question, data, a model with a baseline model, its limits and a talk.
	**Executive summary**	A short opening paragraph with the answer, the evidence, the main limit and the decision you ask for. A busy reader may read only this.
	**Limitation**	Something your result cannot tell you, such as who the data left out. You say it before you show the result.
	**Portfolio**	A small set of your projects, for example GitHub repositories with a README, that a stranger can open and judge.
	**Rubric**	A marking guide that lists what good work looks like for each criterion. You use it to score your partner.

Tell a result as a story in three parts: the answer, the evidence and the limits



The same story works for any capstone. Put the answer first, so the room knows what the evidence is for.

The evidence behind the loan story, with the baseline model beside it

Measure	Number	What it means
Loans, and defaults	8,000; 1,004	12.6% of loans defaulted.
Held-out test loans	2,000; 251 defaults	Loans the model never saw.
Baseline model: always repaid	87.5% accurate	It finds no defaults at all.
Model, decision threshold 0.5	89.2% accurate	Finds 51 of 251 defaults (20.3%).
Model, decision threshold 0.2	Flags 278 loans	Finds 108 of 251 defaults (43.0%).
Area under the curve (AUC)	0.756	How well it ranks risky above safe: 0.5 is a coin flip.

Same loans and the same split as Sessions 14 to 17 (logistic regression, 27 columns, decision threshold 0.2). Your Session 16 numbers may differ a little if your model or decision threshold differ.

An accuracy number alone misleads: show the baseline model beside it

The number alone

The model is 89.2% accurate.

It sounds strong.

With the baseline

Always saying repaid is 87.5% accurate.

At decision threshold 0.5 the model finds 51 of 251 defaults.

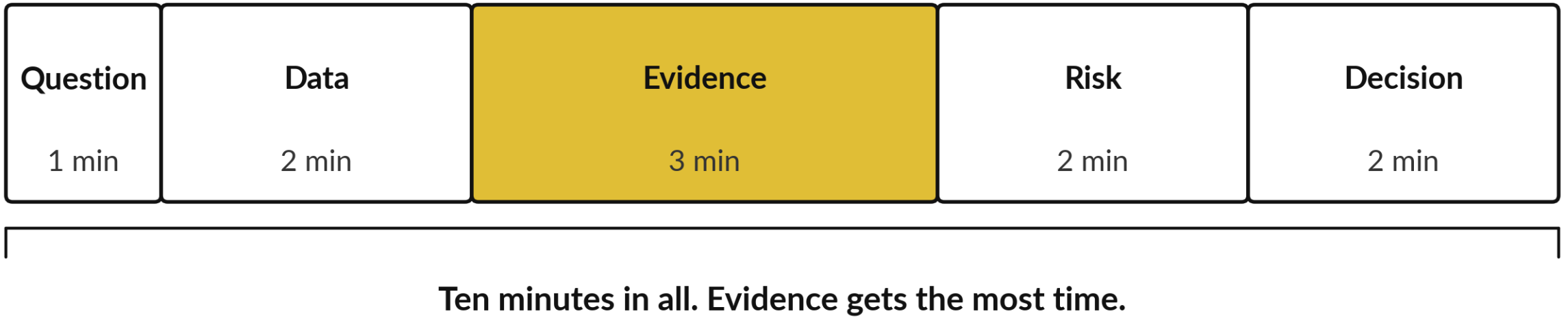
Report the baseline, the share of defaults found and the limit in one sentence.

An executive summary gives the answer, evidence, limit and request in four sentences

An executive summary is the short opening paragraph that a busy reader may read alone. Illustrative example for the loan model:

Answer	Use the model to choose which loans a loan officer reviews first.
Evidence	On 2,000 held-out loans it found 43% of defaults by flagging 14% of loans. The baseline model finds none.
Limit	The data is fictional and 379 loans lack income. We have not tested it after the data changes.
Request	Approve a three month trial where a person decides, with a fairness check by gender each month.

A ten-minute talk has five slides: question, data, evidence, risk and decision



Risk: what could go wrong, and who checks. Decision: what you recommend.

Presenting to a manager: say the real term, then what it means for a decision

	Term	What you say next
	**Recall (sensitivity)**	Of every 100 loans that default, how many we flag.
	**Precision**	Of every 100 loans we flag, how many really default.
	**Area under the curve (AUC)**	How well the model ranks risky loans above safe ones. 0.5 is a coin flip.
	**Baseline model**	The simple rule the model has to beat.
	**Decision threshold**	The risk level at which a loan goes to an officer.
	**Test set**	Loans the model never saw while learning.

Start with the answer. One number per slide. Keep the notebook ready for questions.

Say the limits before the result, and the room trusts the result

Result first

Opens with the best number.

Limits arrive in the last minute.

The claim lands before the caveat.

Limits first

Opens with what the data covers.

Names what the model cannot do.

Then shows the result.

A limit you state loses no marks. A hidden one does.

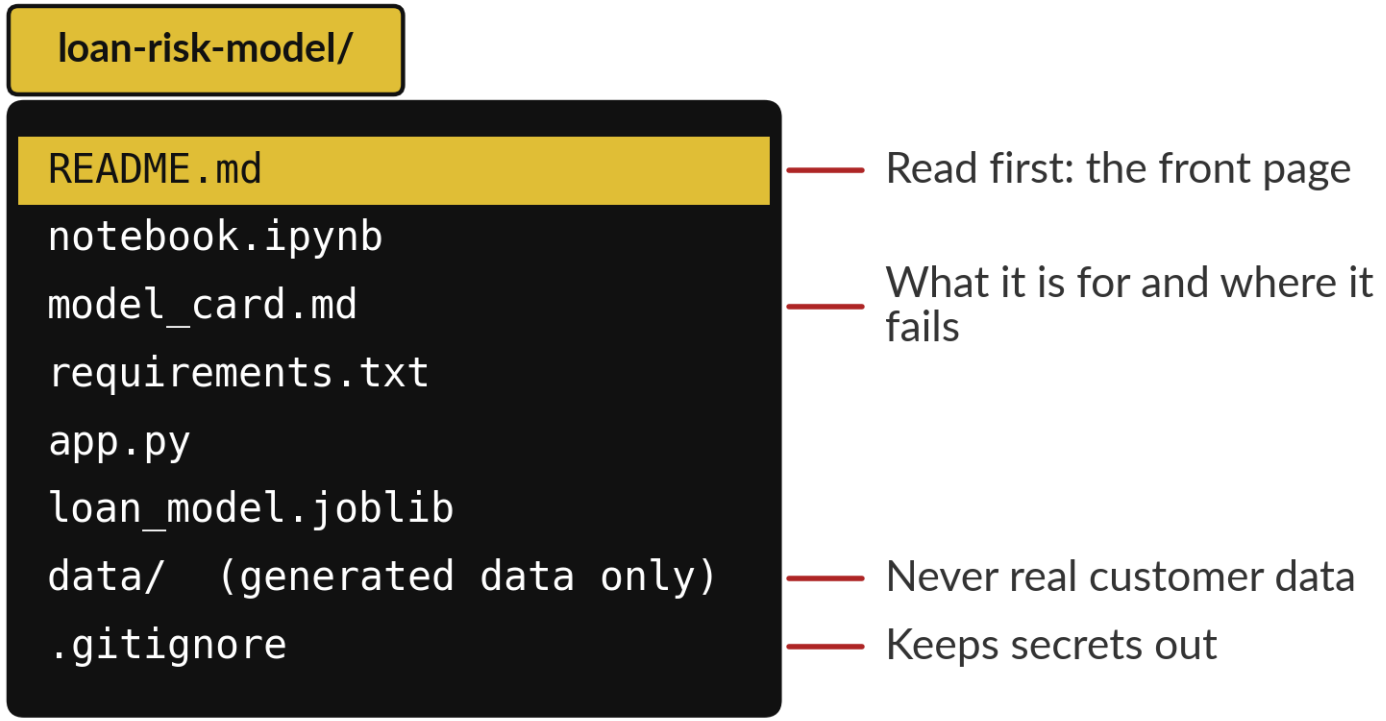
Three questions on method, and answers you can give

Likely question	A good answer
How do you know it works on new loans?	I kept 2,000 loans the model never saw as a test set, and compared the result with a baseline model.
Why that metric and that decision threshold?	Missing a default costs more than reviewing a good loan, so I watch recall. At decision threshold 0.2 it finds 43% of defaults, and about 39 of every 100 flagged loans really default.
Could this result be luck?	This was one split. I would repeat it with k-fold cross-validation and report the range of scores.

Three questions on risk: drift, fairness and data protection

Likely question	A good answer
What could go wrong once people use it?	The data could drift after training. I would compare each month of inputs with the training period, compare predicted with actual defaults, and set a review date.
Is it fair, and are we allowed to use this data?	I audited approval and error rates by gender: 86.5% of women and 85.6% of men approved. I checked the purpose and data minimisation against the Data Protection and Privacy Act, 2019 (a teaching summary, not legal advice). A person decides.
Who is responsible for it?	(Example answer for our fictional lender.) The credit manager owns the model card, and the data protection officer reviews each release.

A portfolio on GitHub is a few repositories a stranger can open in five minutes



Portfolio: a small set of your projects that others can open and judge.

Two or three projects you can explain well.

Each notebook runs from the top.

Never real customer data.

A README answers four questions: what, result, limits and how to run

Section	Loan model example
The question	Which loans should an officer review first?
The result	At decision threshold 0.2 the model finds 43% of defaults on 2,000 held-out loans. The baseline model finds none.
The limits	Fictional data. Not tested after the data changes. Fairness checked by gender only.
How to run	Install requirements.txt, then Runtime, Restart session and run all.

Write it for a stranger. Say which data is generated.

A rubric marks four things, and you score your partner with the same four rows

	Criterion	Strong work (3)	Partly (2)	Not yet (1)
	**Question and evidence**	A one-sentence question. Every claim points to data.	A question, but some claims lack data.	No clear question, or claims without data.
	**Metric and baseline**	A metric chosen for the cost, on unseen data, beside a baseline model.	A metric, but no baseline.	One accuracy number, no baseline.
	**Limits and fairness**	Limits first. Risks named. Groups compared. Data duties checked.	Some limits, said late.	Limits missing or hidden.
	**Clarity**	Five slides, ten minutes, plain words, one decision.	Mostly clear, a little over time.	Over time, jargon, no decision.

The lab asks for one comment on each row.

What this course did not cover: an honest gap list

Area	What that means for you
Mathematics by hand	You calculated results by hand but did not prove them.
Data structures and algorithms	Interview puzzles need separate practice.
Big data tools	Your data fitted in one notebook.
Paid cloud platforms	AWS, Azure, Google Cloud. You used free tiers of Colab, GitHub and Streamlit.
Deep learning from scratch	One neuron and a small network in numpy. The pretrained model was optional.
Dashboards and BI tools	Power BI and Tableau were named, not taught.
Production MLOps	No container, automated tests or monitoring.

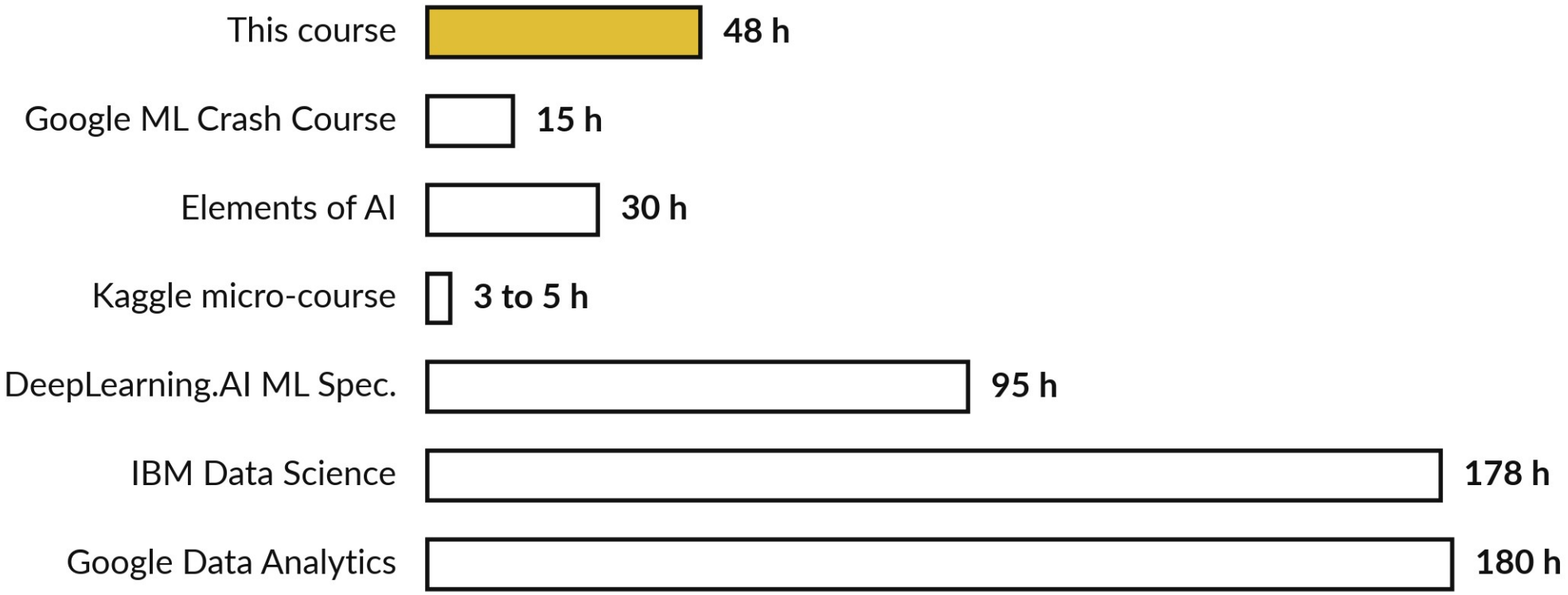
Say these gaps yourself, before anyone asks.

Free next steps by goal: check the current terms before you start

Goal	Free next step
Python habits	Harvard CS50P (Introduction to Programming with Python)
Everyday data work	Kaggle Learn: Pandas
Models beyond this course	Kaggle Learn: Intermediate Machine Learning
Machine learning basics again	Google Machine Learning Crash Course
Deep learning	fast.ai Practical Deep Learning for Coders
AI ideas and ethics	University of Helsinki: Elements of AI
Language models	DeepLearning.AI short courses
Cloud, big data, dashboards, MLOps	Search for a free introductory course on the tool your employer uses. Check its terms.

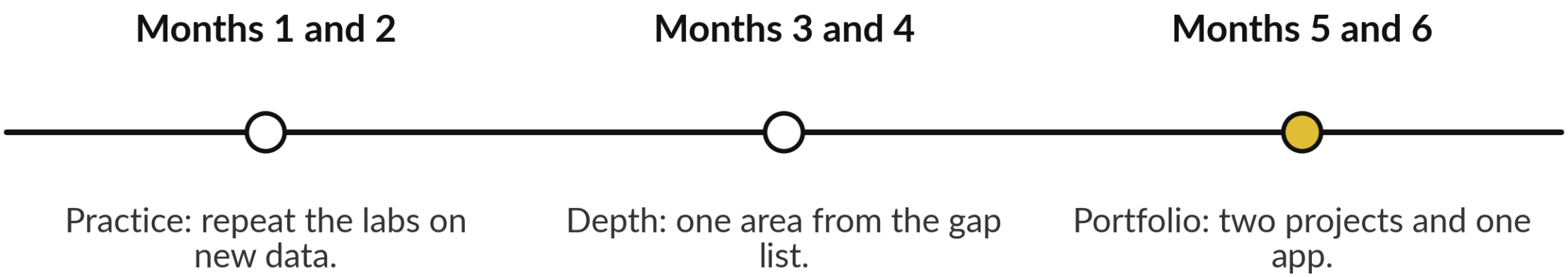
Free to learn, but audit, certificate and account rules change. Check each provider page.

This course is 48 hours. Well-known programmes list about 3 to 180 hours



Provider estimates at the time of writing. Some are marketing figures. They change.

Your six month path: practice, depth, portfolio, at 6 hours a week



6 h a week for 26 weeks is 156 h.

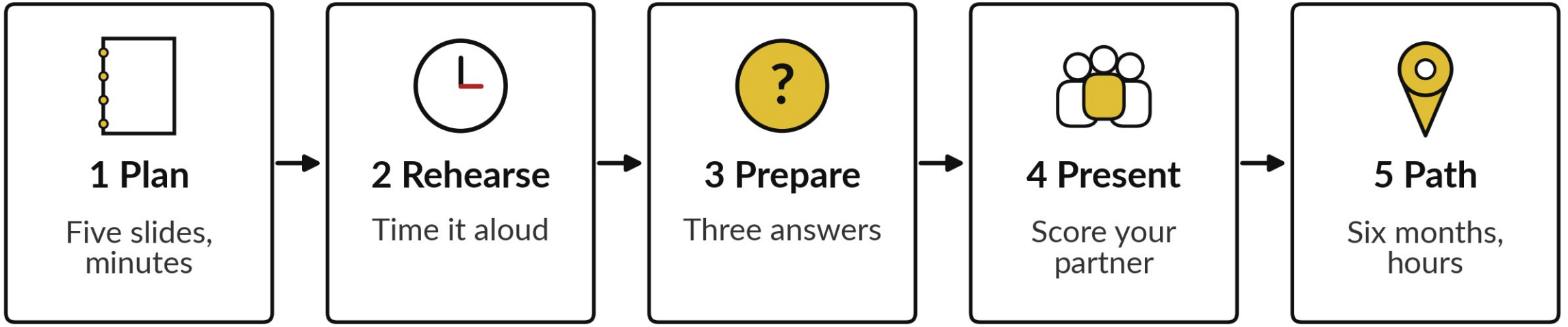
Where you will use this at work: present a result so a manager can decide

Your job	The presentation	What today lets you do
Credit analyst at a bank or SACCO	A monthly credit committee	Open with the answer and the baseline model, say the limits first, and show approval rates by group.
Officer in a ministry	A briefing note for a permanent secretary	Write the executive summary in four sentences, with one number and its source.
Data officer at an NGO	A donor report	State what the data covers and leaves out, and keep a portfolio of reproducible work.
Analyst at a retailer	A branch managers meeting	Explain recall and precision in plain words, with the cost of a mistake in shillings.

Part 2: rehearsal and feedback

Close the slides. Open the sheet.

The lab has five parts, and you keep a presentation sheet



Six tasks in 45 minutes. For each task, plan, do it, then check your numbers.

Your feedback tab: four rubric rows, a score from 1 to 3, and one comment on each

	A	B	C
	Criterion	**Score**	**One comment**
1	Question and evidence	3	Keep: one clear question on slide 1
2	Metric and baseline	2	Add the baseline beside the number
3	Limits and fairness	3	Say the limit before the result
4	Clarity	3	Finished in 10 minutes
5	Total	=SUM(B2:B5) gives 11	Out of 12

Illustrative scores for the formula check. Your partner fills this in for your talk.

Check your numbers before you move on

	What you compute	You should get
	**Planned minutes (task 1)**	1 + 2 + 3 + 2 + 2 = 10
	**Rehearsal total, in seconds and minutes**	710 s = 11 min 50 s
	**Slide that ran longest past its plan**	Evidence, 60 s over
	**Recall at decision threshold 0.2 (task 3)**	108 / 251 = 43.0%
	**Precision at decision threshold 0.2**	108 / 278 = 38.8%
	**Peer scores 3, 2, 3, 3: total and mean**	11 and 2.75
	**Learning path hours**	6 x 26 = 156

If your number differs, find the task that went wrong.

Five likely mistakes in the lab and how to fix them

What you see	What it means	What to do
In the minutes table: 11 where you expected 10	The minutes add up to more than the plan.	Cut a sentence from the longest slide, then add the minutes again.
#VALUE! in a time formula	Text where a number is expected, such as "70 s".	Type 70 only. Put the unit in the header.
In the feedback tab: 12 where 11 was expected	The range includes the total row, or the header row.	Use B2:B5 for four scores.
Everything scored 3	The scores are not separating strong from weak.	Find one thing to improve on each row.
A path with no named course	Without a course and a date, it stays a wish.	Name one course, a start date and the hours a week.

Exit check: three questions, then compare with the short answers

	Question	Short answer
	**Which three parts make a result into a story?**	The answer, the evidence beside a baseline model, and the limits.
	**Name two things to say before you show a result.**	What the data covers and leaves out, and what the model cannot do. Name the baseline too.
	**Where do you confirm the terms of a free course?**	On the provider page, before you start, because free audit and certificate rules change.

Save your sheet, and start your six month path this week

What you keep from today

Keep your presentation sheet: the five-slide plan with minutes, your rehearsal times, three prepared answers, your partner scores and your six month learning path.

Take away

Your lab file

A talk plan

Your partner

Scored on 4 rows

What next

A 6-month path

What happens next

There is no Session 25. Your six month path starts now. Tell your tutor your first free course and the day you will start.